

What Did Ten Points of Mean Recall Buy? Separating Group-Level from Case-Level Gains in Scene Graph Generation

Anonymous CVPR submission

Paper ID ****

Two frozen models, one cell: relabelling within ϕ removes case identity and nothing else

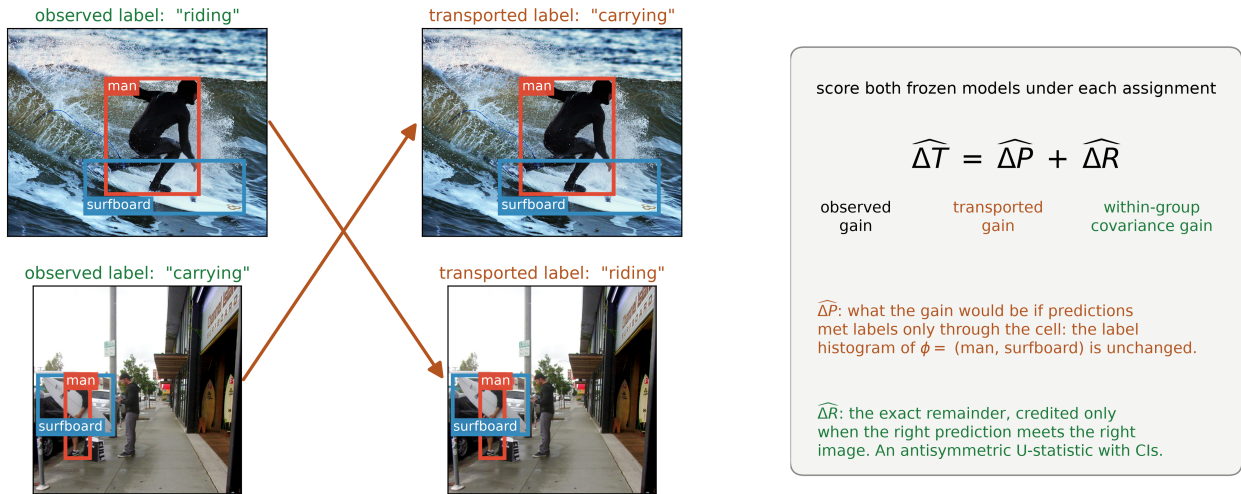


Figure 1. **Within-group relabelling.** Two Visual Genome relations share the subject–object pair (man, surfboard). Swapping their labels keeps the pair’s predicate histogram exactly as it was but breaks the link between each image and its own label. Scoring two frozen models under the observed and the swapped labels splits their score gain exactly: $\widehat{\Delta P}$ is the part that survives the swap — earned at the level of the group — and $\widehat{\Delta R}$ is the part that needs the right label on the right image.

Abstract

001 Debiasing methods for scene graph generation are judged
002 by mean recall, and one of the best known, Total Direct Ef-
003 fect (TDE), raises it on Visual Genome from 14.6 to 24.8.
004 What did those ten points buy? A model raises mean re-
005 call either by predicting rare predicates more often within
006 each subject–object pair, or by recognising which individ-
007 ual images actually show them, and the metric credits both
008 alike. We give an exact way to tell them apart. Scoring
009 each prediction against the label of a different relation from
010 the same subject–object group keeps the group’s predicate
011 frequencies but breaks the link between each relation and
012 its own label; this splits the difference between two frozen
013 models, in a proper score or in mean recall itself, into a

group-level part and a case-level covariance, with no fitted baseline, in $O(NK)$ time, with image-clustered intervals. On TDE’s released checkpoint, 87% of the mean-recall gain is group-level and almost none of it comes from the fifteen rarest predicates. A one-line logit adjustment of the baseline, which leaves the ranking of relations within every group unchanged, recovers nine of the ten points with the same split, at under a tenth of TDE’s cost in recall, and a threshold-free within-group AUC finds no discrimination that TDE gained. The converse also occurs: a CLIP model reading image crops, ranked last on accuracy and on the proper score, discriminates better among relations sharing an object pair than the geometry model ranked above it.

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1. Introduction

Scene graph generation is evaluated on a benchmark whose labels are unusually predictable from something every model sees. Given the ordered object pair (man, surfboard), the predicate on Visual Genome [15] is very often riding, and a lookup table of training frequencies is famously competitive [36]. Recall@ K rewards models that echo this prior, so the field now leads with mean recall, which averages recall over predicates and so rewards getting the rare ones [4, 31]. Debiasing methods are now judged by how far they raise it. Total Direct Effect (TDE) [32], among the most cited of them, raises mean recall@50 of a MOTIFS model from 14.6 to 24.8 on the canonical VG150 split, in our evaluation of the released checkpoint with the authors' own code.

What did those ten points buy? There are two ways to raise mean recall, and the metric cannot tell them apart. A model can move probability towards rare predicates *within each subject-object pair* — say carrying more often for (man, surfboard) — without becoming any better at deciding *which* images show a man carrying rather than riding. Or it can learn to make that decision: put more probability on the correct predicate for the individual image, beyond what the object pair implies. The first is a change in the model's group-level predictions; only the second is better recognition. Every existing remedy — head/body/tail-stratified recall [18], independent mean recall [20], zero-shot recall [22], rarity-stratified accuracy [14] — slices one model's performance differently. None attributes the gain *between* two models to one route or the other.

An exact split. We separate the two routes with one operation (Fig. 1). Take two frozen models and a test set grouped by a discrete feature both models observe, here the subject-object pair. Score each model's prediction for one relation against the label of a *different* relation in the same group. This relabelling preserves each group's predicate frequencies exactly, so any part of the score gain earned at the level of the group survives it; it destroys the correspondence between an image and its own label, so any part earned by getting individual cases right does not. The observed gain therefore splits exactly into a *group-level* part $\widehat{\Delta P}$ and a *case-level* remainder $\widehat{\Delta R}$, which is a within-group covariance between the change in the models' predictions and the label (Theorem 1). The split holds in the observed sample for any proper score and for mean recall itself, needs no fitted reference model, held-out fold or tuning parameter, costs $O(NK)$ on cached predictions, and comes with image-clustered intervals.

What it finds. We re-ran TDE's own released checkpoint with the authors' code and replayed the official evaluator

exactly, so that recall can be computed under any labelling, and split mean recall itself (Sec. 5). Of the ten points, 87% is group-level, the body of the predicate distribution gains, the head loses, and the fifteen rarest predicates contribute almost nothing: TDE recalls one of them, at 0.016. Recall of parked on rises from 0 to 0.886 because TDE now says parked on for (car, street). The case-level remainder needs care, and handling it is part of the method: a thresholded metric earns case-level credit whenever its operating point moves. Subtracting the training predicate prior from the baseline's logits [24], which leaves every within-group ranking exactly as it was, recovers nine of the ten points with the same split, at under a tenth of TDE's cost in $R@50$; against it, TDE's case-level part is -0.0008 with an interval straddling zero, and a threshold-free within-group AUC finds no discrimination TDE gained. The converse appears too. A model reading image crops through a frozen CLIP encoder [26], which every aggregate measure ranks last, gains $+0.00467$ of case-level covariance over the geometry-based model ranked above it, with calibration matched and an interval excluding zero (Sec. 6): its deficit is entirely group-level. That gain does not make it the better ranker, though, and the two measures' disagreement is itself part of what Sec. 6 reports.

Contributions.

- An audit of a published debiasing method's released checkpoint that splits its ten-point mean-recall gain exactly: 87% group-level, a case-level part no larger than a ranking-preserving logit adjustment's, and no gain in within-group discrimination — with a reversal in which the model ranked last is the best discriminator among relations sharing an object pair.
- An exact, model-free decomposition of the difference between two frozen models into group-level and case-level parts, for every Bregman score and for label-weighted hit rates such as mean recall, computed in one pass over cached predictions, with cluster-robust inference, and paired with an operating-point control and a threshold-free AUC.
- Validation that the case-level part is exactly zero when it should be, covers at the nominal rate under image clustering, and separates calibration, prior and shortcut effects — together with a protocol for the one confound it does not separate.

What is not claimed. The split is conditional on the declared grouping: $\widehat{\Delta R}$ measures case-level discrimination *among relations sharing an object pair*, and credits any signal varying inside a pair, including a shortcut nobody recorded. It is not invariant to recalibration — rescaling a model's confidence moves score between the two parts

while changing no ranking — which is why the comparisons whose models differ visibly in confidence, TDE’s and the CLIP model’s, are read confidence-matched. For mean recall the split is of the micro-averaged version, which pools relations across images; it tracks the official per-image average but is not it. And the proper-score split’s headline holds at the configuration we argue for, not at every configuration; Appendix O reports all of them. These limits are taken up in Sec. 7.

2. Related work

Bias and its evaluation in scene graph generation. Scene graph generation was posed as structured prediction over a whole image, solved by message passing between object and relation nodes [34]; MOTIFS then showed a frequency predictor of the predicate given the object pair to be surprisingly competitive [36]. Mean recall became the headline metric, and the causal TDE intervention was introduced, to reduce that dependence [32]; a long line of debiasing work has followed [16, 23], and the metrics themselves have been shown to reward pathologies [20], a diagnosis that recurs in open-vocabulary settings [19]. Benchmark audits outside scene graphs find the same pattern: VQA models collapse when answer priors change between train and test [1], GQA-OOD stratifies accuracy by answer rarity within question groups [14], and multimodal models exploit non-visual shortcuts [10, 37]. Long-tailed recognition faces the same question for re-weighting and logit adjustment [3, 24]. All of these characterise a *single* model. We attribute the gain *between* two.

Proper-score decompositions. Conditional on a grouping, a proper score splits into reliability and resolution terms [7, 25], for general proper scores [2]. Our case-level term is not that resolution term: it descends from the *covariance* decomposition of the Brier score due to Yates [35], which a monotone recalibration moves while the classical resolution is invariant to it. The two coincide only when both models are calibrated within every group. Variance estimators exist for single-model decompositions [30], but not for a contrast between two models scored on the same data.

Comparing two predictors. Inference on a paired score difference is the Diebold–Mariano test [9] and its conditional form [11]; at a single group our total-gain interval is a clustered Diebold–Mariano test (Corollary 3), so the decomposition adds a split to what those tests already report. DeLong *et al.* [8] compare correlated ROC areas with a paired U-statistic, the same device we use for a group-conditional score gain. The case-level term is an order-two U-statistic [13], with a Hájek projection supplying its influence function [29].

3. Splitting a score gain

Setup. Let $q_0(\cdot | x)$ and $q_1(\cdot | x)$ be the predictive distributions of an old and a new frozen model over K labels, and $S(q, y)$ a bounded proper score, larger being better. Our default is the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, negative Brier loss up to a label-only constant; the log score is the second case we report. Let $C = \phi(X)$ be a discrete grouping both models observe, declared before the audit — for relation prediction, the ordered subject–object class pair. We call a value of C a *cell*. The gain vector

$$H_x(y) = S(q_1(\cdot | x), y) - S(q_0(\cdot | x), y), \quad y = 1, \dots, K, \quad (1)$$

can be evaluated at a counterfactual label without rerunning either model, and that is all the construction needs.

Transport. For a cell c , let $\Delta T_c = \mathbb{E}[H_X(Y) | C = c]$ be the observed gain and $\Delta P_c = \mathbb{E}_{H|c} \mathbb{E}_{Y'|c}[H_X(Y')]$ the gain when each case is scored against an *independent* label Y' drawn from the same cell, and set $\Delta R_c = \Delta T_c - \Delta P_c$. Transport keeps the cell and its label distribution and removes only the pairing between a case and its own label.

Theorem 1 (Transport and covariance identities). *For any integrable score contrast H ,*

$$\Delta T_c = \Delta P_c + \Delta R_c,$$

$$\Delta R_c = \sum_{y=1}^K \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | C = c). \quad (2)$$

For the quadratic score, with $\Delta q_y(X) = q_1(y | X) - q_0(y | X)$,

$$\Delta R_c = 2 \sum_{y=1}^K \text{Cov}(\Delta q_y(X), \mathbb{1}\{Y = y\} | C = c). \quad (3)$$

Consequently, if the gain vector is a function of c alone, $\Delta R_c = 0$.

Eq. (3) is what the split means. The new model earns case-level credit only insofar as its probability *change* rises on the label the individual case actually carries, relative to other cases in the same cell. Anything constant within a cell — shifting probability towards rare predicates for every (man, surfboard) relation alike — lands in ΔP , which is why we call it the group-level part. It is *not* simply “prior fit”: under the quadratic score ΔP_c is the improvement in how close the model’s mean forecast sits to the cell’s label frequencies *minus* the increase in the dispersion of its forecasts (Proposition 1). The covariance form is due to Yates [35] and generalises: for every Bregman score, including the log score, ΔR_c is the same covariance with Δq_y replaced by the contrast of the y -th gradient coordinate (Theorem 2). All proofs are in the supplementary.

Why it matters which part a gain is in. One cell, four cases, three labelled 1 and one labelled 2; the old model predicts $(\frac{1}{2}, \frac{1}{2})$ for all four. Model A predicts $(\frac{3}{4}, \frac{1}{4})$ for every case — the cell’s frequencies. Model B predicts $(\frac{3}{4}, \frac{1}{4})$ for the three label-1 cases and $(\frac{1}{4}, \frac{3}{4})$ for the label-2 case: the same distance from the old model, but pointed at each case’s own label. Under the quadratic score A gains $+\frac{1}{8}$, all of it group-level, and B gains $+\frac{3}{8}$ with a case-level part of $+\frac{1}{2}$. Now let deployment see this cell’s labels in proportions $(\frac{1}{4}, \frac{3}{4})$ instead, a label shift. Reweighting each case by $p'(y)/p(y)$ sends A to $-\frac{3}{8}$ — worse than the model it replaced, with not one prediction changed — and leaves B at $+\frac{3}{8}$. The split said in advance which gain was exposed. With $m_y(z) = \mathbb{E}[H_X(z) \mid Y = y, C = c]$, the case-level part is a sum over label pairs, $\Delta R_c = \sum_{y < z} p(y)p(z) D(y, z)$ with $D(y, z) = m_y(y) - m_y(z) + m_z(z) - m_z(y)$, and a label shift leaves every D fixed and changes only the weights. With two labels, as here, a shift can rescale the case-level part but never reverse it; with three or more it can (Appendix C). A case-level gain is therefore less exposed to a shift than a group-level one, not immune to it. That is the operational reason to want them apart. (Appendix C works the example through.)

Estimator. In a cell with $n_c \geq 2$ cases, write $H_i(y) = H_{x_i}(y)$ and estimate

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_i H_i(y_i), \quad \widehat{\Delta P}_c = \frac{1}{n_c(n_c-1)} \sum_{i \neq j} H_i(y_j), \quad (4)$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j} A_{ij}, \quad (5)$$

$$A_{ij} = \frac{1}{2} \{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \quad (6)$$

These satisfy $\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c$ exactly in every sample, and $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic [13] for ΔR_c (Theorem 3). Transporting a case only to the *other* cases’ labels matters: fitting the cell’s label frequencies on all n_c cases, the natural plug-in, shrinks every cell’s case-level gain by exactly $(n_c - 1)/n_c$, halving it at $n_c = 2$ (Proposition 4). With m_{cy} the count of label y in cell c , case i ’s transported score is

$$P_i = \frac{\sum_y m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}, \quad (7)$$

so the whole estimator costs $O(NK)$ on cached predictions, with no fitted model, held-out fold or smoothing parameter. Cells are weighted by size; singletons cannot identify a within-cell counterfactual and are excluded and reported.

Inference. With $\bar{A}_i = (n_c - 1)^{-1} \sum_{j \neq i} A_{ij}$, the Hájek influence contribution of case i is

$$\widehat{\psi}_i = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}. \quad (8)$$

We sum $\widehat{\psi}_i$ within each image and use the one-way cluster-robust sandwich variance, since relations from one photograph are not independent. The estimator is asymptotically normal under bounded scores and non-dominating clusters (Theorem 4); at a single cell, the total-gain interval is a clustered Diebold–Mariano test [9] (Corollary 3). A randomization test that cyclically shifts labels within every cell is a secondary, relation-level diagnostic.

Confidence matching. One property of the split must be part of the procedure rather than a caveat. Because ΔR is a covariance between probability *movements* and labels, rescaling a fixed model’s confidence moves score between the two parts while changing no ranking, no top-1 decision and no information about any case (Sec. 4 measures how much). Two released checkpoints that differ in calibration therefore cannot be compared on their raw outputs. Whenever compared models differ visibly in confidence we split the images at random, fit each model a single temperature by maximum likelihood on one half, and run the split on the other half with each model’s own temperature applied. The protocol uses no label from the audited half.

Splitting mean recall. Nothing in Theorem 1 or the estimator uses properness: both hold for any contrast $H_x(y)$ that can be evaluated at a counterfactual label. Mean recall@ k is one. Let $h_x^{(m)}(y) \in \{0, 1\}$ say whether model m would recall relation x among its top k if its label were y ; replaying the benchmark’s matching rules on cached scores gives it for every y (Appendix O). Pooling relations, mean recall is $\sum_i w(y_i) h_{x_i}(y_i)$ with $w(y) = 1/(Kn_y)$, so $H_x(y) = Nw(y)[h_x^{(1)}(y) - h_x^{(0)}(y)]$ makes $\widehat{\Delta T}$ its change; transport preserves every n_y , and restricting w to one tier or predicate splits that share exactly. One difference matters for reading the result. A hit is a thresholded decision, so moving a model’s operating point — subtracting a log-prior from its logits, say — flips first the cases it already scored as most like the favoured label, and those carry that label more often: a case-level part with no new discrimination. We therefore read the split against an *operating-point control*, the old model’s logits shifted by the training log-prior [24], and pair it with a threshold-free test, the *within-cell AUC*: the probability that a case labelled y has larger log-odds $\log q(y) - \log q(z)$ than one labelled z in the same cell, averaged with weight $n_c(y)n_c(z)$ as in the pairwise form above. A temperature, and any change constant within a cell, leaves it unchanged.

Choosing the grouping. The split is conditional on ϕ . Merging cells changes it by an explicit between-cell covariance term of either sign (Proposition 2), so the finest grouping that the benchmark’s construction defends is the conser-

vative choice. A relevant variable left out of ϕ is the same term with the fully adjusted cell as the fine partition, and Proposition 3 bounds it by a single elicited correlation, in the spirit of the robustness value of Cinelli and Hazlett [5].

4. Does the split measure what it claims?

A released checkpoint cannot validate the split, because what the model can see is unknown. We validate it where the answer is known, first in simulation and then on Visual Genome predictors whose information content we control. All intervals are 95%.

Simulation. We draw 24 cells over five labels with Dirichlet label distributions, take the old model to be exactly the cell prior, and let the new one interpolate towards an oracle. The estimated case-level gain grows linearly with the injected signal. Under a null that perturbs the prior predictions and reassigns them within cells, the randomization test rejects in 0.030 of 200 datasets at level 0.05, with mean estimate -8.97×10^{-5} . For the interval we use on real data, we simulate relations clustered into images with a shared per-image effect and Zipf-distributed cell sizes, most cells holding two cases: the image-clustered interval covers in 94.0% of 500 runs, against 88.6% for an interval that treats relations as independent.

Three designs check that the two parts separate what they claim to. A *prior-only* improvement, where the new model corrects a noisy estimate of the cell prior, gives a case-level gain centred at zero (0.0002, sd 0.0035). An *unrecorded within-cell shortcut* — a binary feature varying inside each cell that the new model uses — is credited to the case-level part (+0.041, sd 0.004): the split measures all within-cell signal relative to the declared grouping, shortcut or not. And a *calibration-only* change, where the new model is a temperature-softened copy of an overconfident one carrying identical case-level information, is credited a large spurious case-level *loss* of -0.488 on raw outputs, which the confidence-matching protocol of Sec. 3 reduces to -0.0003 (sd 0.0017). That last design is why the protocol is part of the procedure.

Controlled predictors on VG150. We train predicate classifiers whose inputs we control on a VG150-style split we reconstruct from the released Visual Genome annotations, with the same 150-object and 50-predicate vocabulary and images divided 70/30 between fitting and audit (Appendix N.1). It is not the canonical split of Sec. 5, which is why its test set is larger; models trained here from annotations need not reproduce a released split. The predictors are: a frequency table that sees only the class pair (FREQ), an MLP on class embeddings (MLP-CLASS), and MLPs adding twelve box-geometry features (MLP-

SPATIAL, MLP-SPATIAL+). The audit covers 227,337 of 229,605 test relations, 99.0%, in 6,346 identified class-pair cells.

MLP-CLASS beats FREQ by 0.03386 of quadratic score and the split assigns *all* of it to the group-level part: the case-level gain is 0.00000 with a zero-width interval and randomization $p = .724$. That zero is algebraic, not empirical — both models are functions of the class pair alone, so Theorem 1 forces it — and it is the sharpest check that nothing is credited to case-level evidence when none exists. Adding box geometry then yields a case-level gain of 0.01439 (CI [0.01373, 0.01504], $p = .002$) that no aggregate score isolates. Against MLP-CLASS the total gain of 0.00885 even *understates* it, because the group-level part deteriorates by 0.00554: geometry moved the model’s probability towards each relation’s own label within its pair, and made it a slightly worse fit to the cell frequencies at once.

5. What TDE’s ten points bought

Tang *et al.* [32] release a causal MOTIFS PredCIs checkpoint that yields two models from one set of weights: the baseline (MOTIFS-SUM) and its Total Direct Effect counterpart (MOTIFS-TDE). The baseline’s predicate logit is a sum of three terms: a visual term from the union region, a context term, and a frequency prior indexed by the object pair, which in PredCIs is fixed by the subject–object cell. TDE subtracts the same sum evaluated with an averaged context. The visual and frequency terms enter both halves unchanged and cancel, so TDE’s output is the context term minus its averaged counterpart: it removes the frequency prior, constant within a cell, and also the image-specific visual term, which is not. We ran the released checkpoint once with the authors’ code on the canonical VG150 test split, saving every branch’s logits for every object pair, and re-scored them offline with an exact replay of the official evaluator’s matching (Appendix O). The replay reproduces the official numbers to the fourth decimal: the baseline reaches $R@50 = 0.6612$ with $mR@50 = 0.1459$, TDE reaches $mR@50 = 0.2476$ while $R@50$ falls to 0.4588, and without the graph constraint TDE’s mean recall *falls*, from 0.3260 to 0.2981. The audit covers 183,642 relations over 26,446 images in 7,127 subject–object cells; 98.9% of the relations fall in the 5,044 cells holding at least two.

An operating-point control. We add one model that needs no training: the baseline’s logits minus $\tau \log \pi(y)$, the training predicate prior, with $\tau = 1$ [24]. Within a cell this adds the same constant to every log-odds $\log q(y) - \log q(z)$, so it changes no within-cell ranking: whatever case-level part it shows is existing discrimination spent at a new operating point, not new discrimination. It reaches $mR@50 = 0.2343$ at $R@50 = 0.6437$.

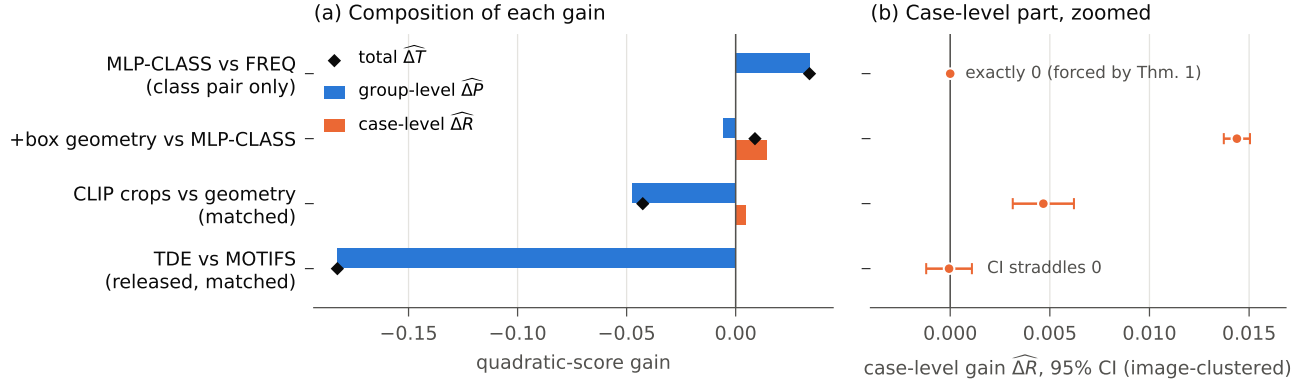


Figure 2. **Where each proper-score gain in this paper lives**, under the quadratic score. (a) Every comparison’s total gain $\widehat{\Delta T}$ (diamond) split into its group-level part $\widehat{\Delta P}$ and case-level part $\widehat{\Delta R}$; the two parts sum to the total exactly. (b) The case-level part alone with its image-clustered 95% interval. Two models built from the class pair alone differ by a purely group-level gain, as Theorem 1 forces. Box geometry and CLIP crops each buy a significant case-level gain, the CLIP model despite losing on the total. TDE’s large proper-score change is group-level, with a case-level part whose interval straddles zero. The CLIP and TDE rows are confidence-matched on held-out images.

Table 1. **Mean recall@50 split** (graph constraint, micro-averaged over identified relations; official $mR@50$ in the first column). LA is the logit-adjusted baseline. The case-level interval is image-clustered. ΔAUC is the change in within-cell AUC, which no calibration or group-level shift can move (LA’s residue is float32 rounding); its interval is an image bootstrap.

	mR	ΔmR	group	case, 95% CI	ΔAUC , CI
TDE—base	.2476	+.0972	+.0841	+.0130 [+.0093, +.0168]	+.0026 [−.0028, +.0086]
LA—base	.2343	+.0941	+.0802	+.0139 [+.0101, +.0176]	+.0001 [−.0000, +.0001]
TDE—LA		+.0031	+.0039	−.0008 [−.0050, +.0033]	+.0026 [−.0029, +.0086]

Mean recall, split. Pooled over identified relations, mean recall@50 rises by 0.0972 (0.1017 on the official per-image average). Of that, +0.0841 is group-level and +0.0130 case-level, interval [+0.0093, +0.0168] (Tab. 1): 87% of the gain survives shuffling every label among the relations that share its object pair. It lives in the body of the distribution. Splitting predicates by training frequency (9 head above 10,000 instances, 26 body, 15 tail below 500), the body gains +0.1224, of which +0.1082 is group-level; the head loses 0.0255; the tail gains 0.0003. Of the 15 tail predicates, 1,480 test relations, the baseline recalls none and TDE recalls one, to, at 0.016: none of the ten points comes from the rarest predicates. Figure 3 shows the mechanism one predicate at a time. Recall of parked on rises from 0.000 to 0.886, almost all of it group-level: TDE now says parked on for (car, street), and so recalls most relations that carry it. Recall of on falls by 0.558 in the same way.

The case-level part is matched by the control. The control reproduces the whole picture: +0.0941 of mean recall, +0.0802 group-level and +0.0139 case-level. Against it, TDE’s case-level part is −0.0008, interval [−0.0050, +0.0033]. So a case-level part of a thresholded metric is not by itself evidence of new discrimination. When the boundary between on and parked on moves, the cases that cross it are those the baseline already scored as more parked on-like, and they are more often labelled so; the covariance of hit gains with labels is real, but it was bought with discrimination the baseline had. The direct test is the within-cell AUC, which no prior shift or recalibration can move: TDE’s change is +0.0026, interval [−0.0028, +0.0086].

The proper score agrees. With each model’s temperature fitted on held-out images ($T^* = 2.50$ baseline, 0.92 TDE), TDE’s quadratic-score change is −0.18264, of which −0.18258 is group-level; the case-level part is −0.00006, interval [−0.00120, +0.00109] (Fig. 2). Under the log score it is +0.04396 against a group-level −0.59157. Against the control the two scores disagree in sign (−0.0040 quadratic, +0.0154 log, both intervals excluding zero), and the control’s own case-level parts (+0.0039, +0.0285) show that the proper-score split, too, responds to the operating point (Appendix O). Unmatched outputs and a coarser grouping give different answers; all configurations are in Appendix O.

What that licenses. On this checkpoint, the ten points of mean recall are a change in which predicates the model favours within each object pair. Everything they did — the

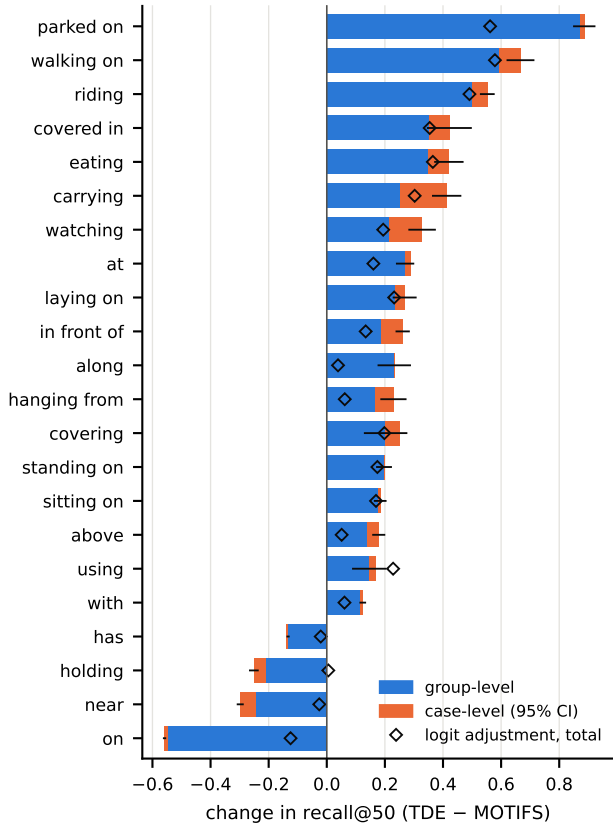


Figure 3. **Where TDE’s mean recall went**, for every predicate whose recall@50 moved by at least 0.12: group-level part (blue), case-level part (orange) with its image-clustered 95% interval, and the total under logit adjustment of the baseline (diamond), which changes no within-cell ranking.

gain, its split, its tier profile — a one-line prior shift that ranks the relations of every cell exactly as the baseline did also does, at under a tenth of TDE’s cost in $R@50$ (0.0175 against 0.2023). Measured against that control, neither recall nor the threshold-free AUC finds any within-cell discrimination TDE gained, and the two proper scores split on its sign.

Why. TDE’s change from the baseline is three subtractions, and splitting along them answers why; the parts add up to TDE’s exactly (Appendix 0.1). Removing the frequency prior, constant within every cell, moves mean recall by -0.0009 and the within-cell AUC not at all. Removing the visual term adds $+0.0164$, with a case-level part of $+0.0016$, interval $[-0.0006, +0.0037]$: within a cell, the context term already knew what the visual term did. Subtracting the averaged context supplies $+0.0817$ of the 0.0972. That term is, to 99.96% of its energy, one vector shared by every relation in every image, correlated $+0.75$ with the log training prior. In PredCls, TDE’s counterfac-

tual is to first order a learned logit adjustment of the context branch, which is why the explicit one reproduces it.

6. The model every metric ranks last

The predictors of Sec. 4 never see an image: they read the class pair and, at most, box coordinates from the annotation. On the same reconstructed split we add one that reads pixels — subject and object regions cropped from the image and encoded by a frozen CLIP ViT-B/32 [26], concatenated with the class embeddings MLP-CLASS uses. CLIP is not fine-tuned, so any case-level gain reflects what pretraining had already recovered. All three compared models are retrained on one fixed subsample of 100,000 training relations, so differences isolate the features rather than the amount of data.

By every aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6161 against 0.6467 for the geometry model, and a quadratic score 0.04655 lower. An evaluator reading either number would discard it. The split disagrees. With each model’s confidence matched on held-out images, the CLIP model loses 0.04727 of group-level score against the geometry model while *gaining* 0.00467 of case-level score (CI $[+0.00314, +0.00621]$); on raw outputs the figures are 0.05296 and 0.00641 (CI $[0.00514, 0.00769]$, $p = .002$). Its whole deficit is group-level (Fig. 2). Against the class-only model, pixels buy 0.01665 of case-level score on raw outputs where box geometry buys 0.01023, with well-separated intervals.

A larger case-level part is not the same as better ranking, and here the two disagree. With each model’s prior matched within cells, the CLIP model still ranks the correct predicate first less often than the geometry model: top-1 accuracy 2.17 points lower, MRR 1.68 lower, recall@5 1.05 lower. Its probability moves towards the correct predicate within a pair more than the geometry model’s does, but not by enough, often enough, to overtake it. So the CLIP model carries more case-level covariance without being the better ranker, and a calibration-free within-cell rank contrast is what would decide which describes recognition.

Two checks support this reading. Post-hoc correction of a model’s label prior should move only the group-level part, and it does: $+0.02401$ group-level against $+0.00111$ case-level. Applied to the CLIP model it halves the aggregate deficit, from -0.04655 to -0.02143 , while the case-level advantage persists ($+0.00752$; $+0.00663$ with both models corrected).

Beyond scene graphs. Nothing in Sec. 3 is specific to relations, and the supplementary applies the identical estimator to long-tailed image classification on CIFAR-100-LT across seeds and imbalance ratios, with superclasses as the grouping, and to long-tailed text classification. On CIFAR-100-LT two standard re-weighting schedules turn out to have opposite compositions once confidence is matched,

reversing what their raw scores suggest about both. The constraint on the grouping bites here: the fine class label cannot serve as a cell, because it leaves every cell label-homogeneous and forces the case-level part to zero by construction — an artefact that reads exactly like a null result.

7. Limitations and conclusion

Limitations. Four bound what the split can say, and we would rather state them than have them found. *It is conditional on the grouping.* The case-level part measures discrimination among relations sharing an object pair; two equally fine partitions of the same cases can disagree completely about whether a gain is case-level (Appendix K), so declaring ϕ is a modelling decision that the benchmark, not the submitting author, should make before a leaderboard opens. *It credits any within-cell signal*, including shortcuts nobody recorded, and our own bound on that exposure is not reassuring: for the controlled-predictor pair of Sec. 4, an unrecorded covariate correlated at 0.061 with both parts would account for the entire case-level gain (Appendix I). *It is not invariant to recalibration or to the operating point:* the comparisons whose models differ visibly in confidence are read confidence-matched, the case-level part of a thresholded metric is read against a control that moves only the operating point, and the within-cell AUC that is invariant to both gives up the exact additivity that is the point of the construction. *And it needs scores for every label:* most benchmarks archive only ranked top- k predictions, and recovering the rest took a rerun of the released checkpoint. That, not compute, is the obstacle to routine use.

Conclusion. A ten-point mean-recall gain is two different achievements summed together, and the field’s metrics report the sum. Scoring each prediction against another case’s label from the same group separates them exactly, with no fitted reference model and with intervals that respect the clustering of relations within images. Applied to a released debiasing method, the split finds a ten-point mean-recall gain that is 87% group-level, whose case-level remainder a ranking-preserving logit adjustment matches, and behind which no within-group discrimination was gained; applied to a model every aggregate metric ranks last, it finds a case-level gain that the aggregates conceal. We suggest that benchmarks whose labels lean on a feature every model sees report the split beside the headline metric: the declared grouping and its coverage, both signed parts with the case-level interval, the confidence-matched row, and, for a thresholded metric, the operating-point control. Code, the results files behind every number, and the scripts that regenerate the model predictions are in the supplementary material.

References

- [1] Aishwarya Agrawal, Dhruv Batra, Devi Parikh, and Anirudha Kembhavi. Don’t just assume; look and answer: Overcoming priors for visual question answering. In *CVPR*, pages 4971–4980, 2018. 3
- [2] Jochen Bröcker. Reliability, sufficiency, and the decomposition of proper scores. *Quarterly Journal of the Royal Meteorological Society*, 135(643):1512–1519, 2009. 3
- [3] Kaidi Cao, Colin Wei, Adrien Gaidon, Nikos Arechiga, and Tengyu Ma. Learning imbalanced datasets with label-distribution-aware margin loss. In *NeurIPS*, 2019. 3, 10, 18
- [4] Tianshui Chen, Weihao Yu, Riquan Chen, and Liang Lin. Knowledge-embedded routing network for scene graph generation. In *CVPR*, pages 6156–6164, 2019. 2
- [5] Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B*, 82(1):39–67, 2020. 5, 13
- [6] Yin Cui, Menglin Jia, Tsung-Yi Lin, Yang Song, and Serge Belongie. Class-balanced loss based on effective number of samples. In *CVPR*, pages 9268–9277, 2019. 10, 18
- [7] Morris H. DeGroot and Stephen E. Fienberg. The comparison and evaluation of forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 32(1-2):12–22, 1983. 3
- [8] Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: A non-parametric approach. *Biometrics*, 44(3):837–845, 1988. 3, 12
- [9] Francis X Diebold and Roberto S Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. 3, 4
- [10] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020. 3
- [11] Raffaella Giacomini and Halbert White. Tests of conditional predictive ability. *Econometrica*, 74(6):1545–1578, 2006. 3
- [12] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016. 18
- [13] Wassily Hoeffding. A class of statistics with asymptotically normal distribution. *The Annals of Mathematical Statistics*, 19(3):293–325, 1948. 3, 4, 6
- [14] Corentin Kervadec, Grigory Antipov, Moez Baccouche, and Christian Wolf. Roses are red, violets are blue... but should vqa expect them to? In *CVPR*, pages 2775–2784, 2021. 2, 3
- [15] Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A Shamma, et al. Visual genome: Connecting language and vision using crowdsourced dense image annotations. *International Journal of Computer Vision*, 123(1):32–73, 2017. 2, 14
- [16] Peng Kuang, Zhibo Wang, Zhixuan Chu, Jingyi Wang, and Kui Ren. Towards real-world debiasing: Rethink-

- ing evaluation, challenge, and solution. *arXiv preprint arXiv:2405.15240*, 2024. 3
- [17] Ken Lang. Newsweeder: Learning to filter netnews. In *ICML*, pages 331–339, 1995. 21
- [18] Rongjie Li, Songyang Zhang, Bo Wan, and Xuming He. Bipartite graph network with adaptive message passing for unbiased scene graph generation. In *CVPR*, pages 11104–11114, 2021. 2
- [19] Rongjie Li, Songyang Zhang, Dahua Lin, Kai Chen, and Xuming He. From pixels to graphs: Open-vocabulary scene graph generation with vision-language models. In *CVPR*, 2024. 3
- [20] Wei Li, Haiwei Zhang, Qijie Bai, Guoqing Zhao, Ning Jiang, and Xiaojie Yuan. Rethinking the evaluation of unbiased scene graph generation. In *BMVC*, 2022. 2, 3
- [21] Zachary C Lipton, Yu-Xiang Wang, and Alexander Smola. Detecting and correcting for label shift with black box predictors. In *ICML*, pages 3122–3130, 2018. 4
- [22] Cewu Lu, Ranjay Krishna, Michael Bernstein, and Li Fei-Fei. Visual relationship detection with language priors. In *ECCV*, pages 852–869, 2016. 2
- [23] Jiaan Luo, Feng Hong, Qiang Hu, Xiaofeng Cao, Feng Liu, and Jiangchao Yao. Long-tailed recognition with model rebalancing. In *NeurIPS*, 2025. 3
- [24] Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. Long-tail learning via logit adjustment. In *ICLR*, 2021. 2, 3, 4, 5, 20
- [25] Allan H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973. 3
- [26] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *ICML*, pages 8748–8763, 2021. 2, 7, 16
- [27] Paul R. Rosenbaum. *Observational Studies*. Springer, New York, 2 edition, 2002. 5, 13
- [28] Marco Saerens, Patrice Latinne, and Christine Decaestecker. Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure. *Neural Computation*, 14(1):21–41, 2002. 4
- [29] Robert J. Serfling. *Approximation Theorems of Mathematical Statistics*. John Wiley & Sons, New York, 1980. 3, 8, 12
- [30] Stefan Siegert. Variance estimation for brier score decomposition. *Quarterly Journal of the Royal Meteorological Society*, 140(682):1771–1777, 2014. 3
- [31] Kaihua Tang, Hanwang Zhang, Baoyuan Wu, Wenhan Luo, and Wei Liu. Learning to compose dynamic tree structures for visual contexts. In *CVPR*, pages 6612–6621, 2019. 2
- [32] Kaihua Tang, Yulei Niu, Jianqiang Huang, Jiaxin Shi, and Hanwang Zhang. Unbiased scene graph generation from biased training. In *CVPR*, pages 3716–3725, 2020. 2, 3, 5
- [33] Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, Cambridge, UK, 1998. 8, 12
- [34] Danfei Xu, Yuke Zhu, Christopher B Choy, and Li Fei-Fei. Scene graph generation by iterative message passing. In *CVPR*, pages 3097–3106, 2017. 3, 14
- [35] J. Frank Yates. External correspondence: Decompositions of the mean probability score. *Organizational Behavior and Human Performance*, 30(1):132–156, 1982. 3, 13
- [36] Rowan Zellers, Mark Yatskar, Sam Thomson, and Yejin Choi. Neural motifs: Scene graph parsing with global context. In *CVPR*, pages 5831–5840, 2018. 2, 3
- [37] Wenhao Zhou, Faqiang Liu, Hao Zheng, and Rong Zhao. Mitigating data bias and ensuring reliable evaluation of ai models with shortcut hull learning. *Nature Communications*, 16, 2025. 3